Bayesian Integration of Censored Viral Concentration and Ct Data for Pathogen Shedding Dynamics Using Stan

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1 Background and Objective

Accurate estimation of pathogen shedding dynamics is critical for epidemiological surveillance and public health decision-making. Viral shedding datasets are commonly reported using two laboratory measurement formats: viral concentration (e.g., log10 copies/mL) and cycle threshold (Ct) values from RT-PCR assays. Both measurement types are frequently subject to right censoring due to assay detection limits. Viral concentration data are censored below the limit of detection (LOD), while Ct values are censored at upper cycle thresholds (e.g., Ct $\geq$ 40).
The objective of this project is to:

1. Develop and implement a Bayesian hierarchical model in Stan
2. Construct a unified Bayesian framework integrating heterogeneous laboratory measurements
3. Estimate a latent viral load trajectory shared across data source

Rather than treating viral concentration and Ct measurements as separate analytical problems, this framework models them as distinct observation mechanisms of the same underlying viral load trajectory. This structure allows the model to coherently integrate multiple data sources with different scales, censoring mechanisms, and measurement error structures into a unified inferential system.

2 Data Description

2.1 Viral Concentration Dataset

The first dataset consists of longitudinal viral concentration measurements reported in log scale. The key characteristics include:

- Observed viral load values
- Right-censored values at lower limit of detection (LOD)
- Time indexed by days since symptom onset

2.2 Ct Dataset

The second dataset consists of longitudinal Ct values. The key characteristics include:

- Ct inversely related to viral load
- Right-censored at Ct detection limit (e.g., Ct ≥ 40)
- Converted to viral load through calibration relationship

3 Statistical Modeling Framework: Exponential Decay Model

Model Assumptions

The modeling framework relies on several key assumptions:

1. Viral RNA concentration follows exponential decay after symptom onset.
2. Measurement errors for viral load and Ct values are normally distributed.
3. Ct values are linearly related to $\log_{10}$ viral concentration through a calibration relationship.
4. Censoring mechanisms (LOD and Ct limit) are non-informative.

This work develops a Bayesian exponential decay model for pathogen shedding dynamics that integrates both qPCR cycle threshold (Ct) measurements and directly observed viral load data. The primary goal is to improve inference on viral decay trajectories by explicitly modeling the relationship between Ct values and RNA concentrations within a unified Bayesian framework.
The underlying biological assumption is that viral RNA concentration decays exponentially over time after symptom onset. Let t denote time since symptom onset (in days), and let $c(t)$ be the viral concentration at time t on the natural scale (copies/mL). When the concentration of pathogen shed $c(t)$ is subject to exponential decay, it decreases at a rate a_0 proportional to the current value. Define:

$$c(t) = c_0 \cdot e^{-a_0 t}$$

Where c_0 is the concentration of pathogen shed at symptom onset.
Taking logarithm base 10:

$$\log_{10} c(t) = \log_{10} c_0 - \frac{a_0 t}{\log 10}$$

3.1 Viral Load Observation Model

The observed viral load values (on $\log_{10}$ scale) are modeled as:

$$Y_{\text{viral}}(t) \sim \mathcal{N}\left(\mu_{\text{viral}}(t), \sigma^2_{\text{viral}}\right)$$

with mean:

$$\mu_{\text{viral}}(t) = \log_{10} c_0 - \frac{a_0 t}{\log 10}$$

3.2 Ct Value Observation Model

Assuming a standard curve allows for DNA quantification, we model a linear relationship between Ct values and $\log_{10}$viral concentration.

$$\log_{10} c(t) = \beta_0 + \beta_1 \cdot \text{Ct}(t) \quad \Longrightarrow \quad \text{Ct}(t) = \frac{\log_{10} c(t) - \beta_0}{\beta_1}$$

Therefore, the observed Ct values are modeled as:

$$Y_{\mathrm{Ct}}(t) \sim \mathcal{N}\left(\mu_{\mathrm{Ct}}(t), \sigma_{\mathrm{Ct}}^2\right)$$

with mean:

$$\mu_{\mathrm{Ct}}(t) = \frac{\log_{10} c_0 - \frac{a_0 t}{\log 10} - \beta_0}{\beta_1}$$

standard deviation:

$$\sigma_{\mathrm{Ct}}(t) = \frac{\sigma_{\mathrm{viral}}}{|\beta_1|}$$

Unified Data Integration Framework

The proposed model treats viral concentration and Ct measurements as two observation layers linked to a shared latent viral load process $VL(t)$. Each data source contributes information through its own measurement model and censoring mechanism, while inference is performed on the common underlying trajectory.

This hierarchical structure allows heterogeneous data types to be integrated without requiring direct transformation of one dataset into another prior to modeling. Instead, the transformation is embedded within the probabilistic framework, allowing uncertainty to propagate correctly across measurement scales.

3.3 Transformation from Ct to Log Viral Load

Given a Ct value x, we can back-transform to estimated viral load:

$$\log_{10} c(t) = \beta_0 + \beta_1 \cdot x$$

.

3.4 Reparameterize the Ct-Forward Form

Originally, the model was written as $\log_{10} c(t) = \beta_0 + \beta_1 Ct(t)$, which implies $Ct(t) = \frac{\log_{10} c(t) - \beta_0}{\beta_1}$. Writing the model in this form introduces division by the unknown parameter β_1 inside the likelihood, and this will create several computational issues:

1. Posterior Instability: If β_1 is near zero during sampling, then $\frac{1}{\beta_1} \to \infty$, which can produce extreme predicted values and unstable posterior geometry.

2. Strong Parameter Correlation: The likelihood depends on the nonlinear combinations $\frac{\beta_0}{\beta_1}$ and $\frac{1}{\beta_1}$, inducing high posterior dependence and slow MCMC mixing.

3. Artificial Variance Inflation: Small fluctuations in β_1 can produce large swings in predicted $Ct(t)$, leading to an artificially wide posterior predictive distribution that reflects parameterization effects rather than biological variability.

3.5 Reparameterized Ct-Forward Model

Let

$$VL(t) = \log_{10} c(t)$$

Then

$$Ct(t) = -\frac{\beta_0}{\beta_1} + \frac{1}{\beta_1} VL(t)$$

Define

$$\alpha_0 = -\frac{\beta_0}{\beta_1},\ \alpha_1 = \frac{1}{\beta_1}$$

So the Ct-forward model becomes

$$Ct(t) \sim \mathcal{N}\big(\alpha_0 + \alpha_1 VL(t),\ \sigma_{ct}\big)$$

3.6 Advantages of the Reparameterization

The reparameterization offers several computational and inferential advantages. First, it removes division by a model parameter within the likelihood, thereby eliminating a potential source of pathological posterior curvature. Second, the resulting formulation induces a linear Gaussian regression structure for the Ct observation model, which improves numerical stability. Third, the posterior geometry becomes more regular under Hamiltonian Monte Carlo, reducing divergences and improving exploration of the parameter space. This reparameterization also reduces dependence among parameters, leading to more efficient sampling and improved effective sample sizes in Stan. Importantly, this transformation does not alter the underlying statistical model; rather, it represents a change of coordinates in parameter space.
The Ct-forward formulation additionally aligns more closely with the biological measurement process, in which latent viral load undergoes PCR amplification before being observed as a Ct value. Conceptually, this corresponds to the sequence:

$$\text{Viral load} \rightarrow \text{PCR transformation} \rightarrow \text{Ct measurement.}$$

Thus, the Ct-forward parameterization improves computational stability while preserving the scientific interpretation of the model.

4 Stan Implementation

To implement the Bayesian hierarchical model, we employed the No-U-Turn Sampler (NUTS), a variant of Hamiltonian Monte Carlo, configured with four independent chains. Each chain consisted of 2,000 total iterations, including a 1,000-iteration warm-up phase to ensure convergence and efficient exploration of the posterior distribution. The prior distributions were selected to be weakly informative: the observation noise sig_obs followed a $Uniform(0, 10)$ distribution, while the intercepts $a0$ and $c0$ were assigned broad $Normal(0, 100)$ priors. For the cycle threshold (Ct) components, we defined $\alpha_0 \sim Normal(40, 10)$ and $\alpha_1 \sim Normal(-3, 1)$ to reflect typical viral load scales, complemented by an $Exponential(1)$ prior for the error term sig_ct.

Statistical Context

The model employs a hierarchical data-integration approach in which multiple measurement types are linked to a shared latent process. Viral load and Ct observations are treated as conditionally independent given the latent trajectory, but are jointly informative for inference.
By explicitly modeling source-specific likelihoods and censoring mechanisms within a single posterior framework, the approach enables principled integration of heterogeneous surveillance data while preserving measurement-scale uncertainty.

Table 1: Model Parameters and Variable Descriptions

Variable	Description
sig_obs	The standard deviation of the measurement error for the viral load concentration.
sig_ct	The standard deviation of the measurement error for the Ct observations.
a_0, c_0	Regression parameters governing the underlying viral dynamics.
α_0	The baseline Ct value when the log viral load is at the reference level.
α_1	The slope representing the change in Ct cycles per unit increase in log viral load.
c_obs	The observed viral load concentration included in the likelihood calculation
ct_obs	The observed Ct values included in the likelihood calculation
censlim	The limit of detection (LOD) used to account for censored viral load data.

Posterior Diagnostics

Model convergence was assessed using trace plots, effective sample size, and the potential scale reduction factor $\hat{R}$. All parameters demonstrated $\hat{R} < 1.01$, indicating satisfactory convergence across chains. No divergent transitions were observed, suggesting stable Hamiltonian Monte Carlo geometry under the reparameterized Ct-forward formulation.

5 Results

Figures 1 and 2 illustrate the individual shedding trajectories over time for the viral concentration dataset *woelfel2020virological* and the Ct dataset *wang2020fecal*, respectively. These figures display the longitudinal patterns of viral measurements across individuals, highlighting both variability and temporal decline in shedding dynamics.

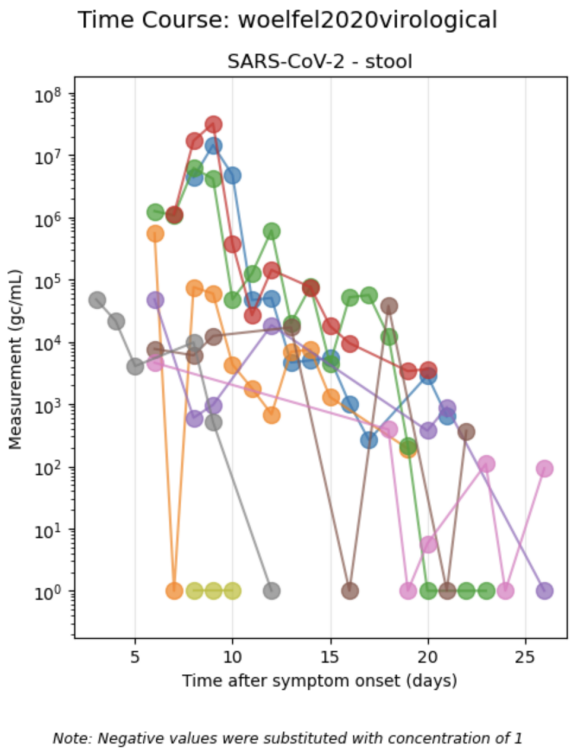

Figure 1: Individual Viral Load Concentration Shedding Trajectories Over Time

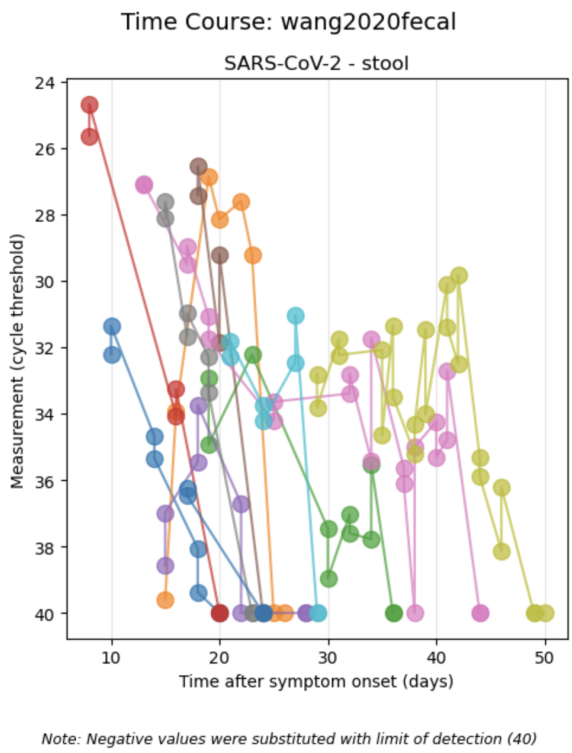

Figure 2: Individual Ct Value Shedding Trajectories Over Time

After incorporating both datasets into the Bayesian hierarchical model, we obtained the combined viral

load trajectory, which integrates observed viral concentration data and Ct-derived viral load estimates into a unified longitudinal framework.

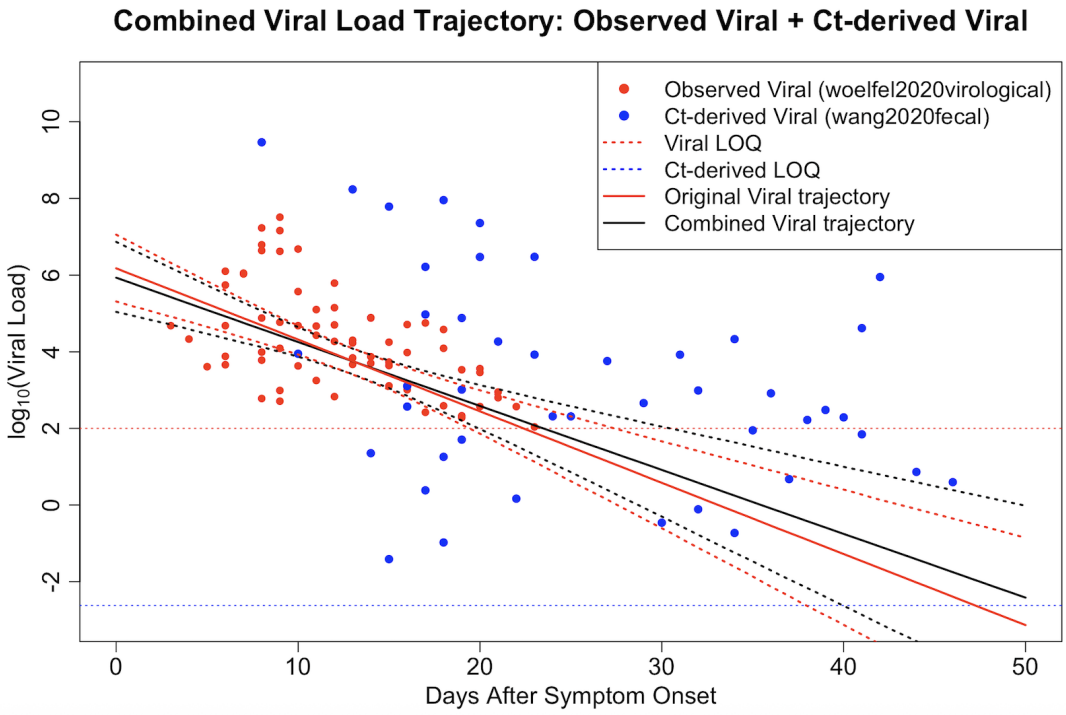

Figure 3: Combined data shedding trajectories

Figure 3 demonstrates the integrated Bayesian estimation of viral shedding dynamics combining observed viral concentration data and Ct-derived viral load estimates. While early-phase trajectories remain consistent across data sources, integration of Ct-derived measurements results in a modest upward adjustment in late-phase viral load estimates. This reflects the additional information contributed by Ct values in periods where viral concentration measurements are sparse or censored. This demonstrates the model's ability to borrow strength across measurement platforms, producing stable inference even when one data source is sparse or heavily censored.

6 Public Health Implications

By serving as a unified data-integration framework, the model enables heterogeneous laboratory measurements to be combined in a statistically coherent manner, improving robustness of inference in real-world surveillance settings where data sources are fragmented. This improves inference on shedding duration and peak infectiousness, which are critical inputs for isolation policy design, transmission modeling, and outbreak surveillance systems. The framework demonstrates how heterogeneous laboratory measurements can be coherently combined to enhance epidemiological decision-making.